



Integrating the PERMA model and SWPBS approach: effects on portuguese early childhood teachers' well-being and children's self-regulation

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ABSTRACT

Early Childhood Education and Care (ECEC) teachers' well-being is crucial for the adoption of educational practices that foster the development of self-regulation skills of children in their classrooms. Promoting these skills in preschool has proven to increase children's school readiness and future academic achievement. Although interventions using Positive Psychology to address teachers' well-being and interventions using School-Wide Positive Behaviour Support (SWPBS) to increase children's self-regulation skills have achieved positive results, these elements have not been addressed simultaneously. In this paper, we measure the longitudinal effects of a one-year intervention, integrating the PERMA model and SWPBS approach into a ten-session training program, on 30 Portuguese ECEC teachers' well-being and 397 children's self-regulation skills. Using repeated measures ANOVA and a Latent Difference Score Model, we found statistically significant positive results between pre- and post-testing on teachers' well-being levels and on children's self-regulation skills. The results indicate that the combination of the PERMA model and the SWPBS approach can increase the effectiveness of interventions and professional development actions aiming to promote the well-being of ECEC teachers and the self-regulation skills of preschool children.

Well-being is a dynamic process that relates to the subjective perception of feeling good and living a purposeful life (Sonnentag, 2015). Its dynamic nature and conceptual complexity encompass individual factors, like physical and psychological health (Hernandez et al., 2018), and external factors, usually associated with the socio-relational environment in which people live (Kansky & Diener, 2017).

One of the environments that influences well-being is the workplace (Haworth, 2004; Stansfeld et al., 2013). Factors like income (Thomson et al., 2022), peer support, autonomy at the job (Hirschle & Gondim, 2020), and the overall social environment at the workplace (Daniels et al., 2017) affect the personal and professional well-being of Early Childhood Education and Care (ECEC) professionals, whose workplace well-being is also significantly influenced by perceptions of relatedness and competence (Jones et al., 2020).

1. The PERMA model for promoting ECEC professionals' well-being

ECEC teachers are required to practice high levels of emotional understanding and management in interactions with children in their classroom (Chang & Davis, 2009). This can accentuate the negative effects that high job demands can have on their well-being over the academic year (Schaack & Stedron, 2020), accumulate with years of service (Muhonen et al., 2024), and potentially contribute to turnover intentions (Heilala et al., 2023). Although research indicates that stress levels are relatively high among these professionals (Smith & Lawrence, 2019) and negatively influence their educational practice (Kokkinos et al., 2005), quality professional development opportunities addressing both stress and teacher-child relationships are still lacking (Sandilos et al., 2018). As such, providing these professionals with improved classroom management techniques and stress coping skills, as well as broader actions to foster teachers' general well-being, is crucial for the quality of ECEC that they provide (Baker et al. 2017; Stone et al., 2023).

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Following evidence indicating that multi-foci, holistic approaches are more effective than targeted interventions (Cann et al., 2024), Positive Psychology has been increasingly used to promote the well-being of ECEC teachers (Shankland & Rosset, 2017; Vo & Allen, 2022). Many of these interventions use the PERMA model (Turner & Theilking, 2019), proposed by Martin Seligman (2012), to promote subjective well-being by working on five elements: Positive Emotions, Engagement, Relationships, Meaning, and Achievement. In practice, using the PERMA model to support ECEC teachers' well-being involves guiding them to foster positive emotions, use character strengths for engagement, build positive relationships, find meaning in daily tasks, and celebrate significant achievements (Falecki et al., 2018). Although empirical research measuring the effects on the participants' well-being levels continues to be insufficient and methodologically dispersed (Cabrera & Donaldson, 2023; de Carvalho et al., 2023; Fox et al., 2023), available evidence indicates that these interventions can achieve positive results not only on teachers' general well-being (Vo & Allen, 2022) but also on their job satisfaction (Dreer, 2024). Thus, they can effectively increase the personal and professional well-being levels of the participants (Yeh & Barrington, 2023).

It is also important to consider that the quality of daily interactions with children and peers is a crucial factor that significantly influences the well-being of ECEC teachers (Ansari et al., 2022; Jones et al., 2020; Stone et al., 2023; Whitaker et al., 2015). While interventions using the PERMA model usually address the importance of establishing and maintaining positive relationships, they do not focus on helping ECEC teachers to deal with reported difficulties in managing challenging behaviour in their classrooms (Gebbie et al., 2012). Since these difficulties can impair the positive effects of interventions solely focused on fostering personal well-being, these could probably benefit from the integration of training to assist ECEC teachers in adopting proactive strategies to deal with discipline issues in the classroom (Chang, 2013).

1.1. The SWPBS Approach for Promoting Children's Self-regulation

School-Wide Positive Behaviour Support (SWPBS) is a proactive behaviour management approach that seeks to prevent the occurrence of challenging behaviour through the adoption of a multi-tiered system organized in three tiers of support. Tier 1 focuses on designing "effective environments that improve the fit or link between research-validated practices and the environments in which teaching and learning occur" (Sugai et al., 2000, pp. 133-134). Tier 1 offers evidence-based, preventive, instructional strategies that use explicit teaching for managing behaviour (Lemos et al., 2023; Sobock & Reister, 2020). Tiers 2 and 3 comprise interventions typically implemented with smaller groups or individual children targeting more intensive behavioral or social needs (Horner et al., 2014). The implementation of SWPBS has achieved positive results in promoting positive behaviours and in increasing the academic achievement of students in elementary, middle, and high school (Michael et al., 2023; Lee & Gage, 2020). Since it has also been effective in promoting teachers' sense of self-efficacy (Ross et al., 2012), it has been increasingly adapted to ECEC contexts (Carter et al., 2011).

Usually described as Program-Wide Positive Behaviour Support (Fox et al., 2015), programs implemented in ECEC settings seek to improve educational practices (Hemmeter et al., 2022). Research shows that these programs can achieve positive results in reducing challenging behaviour (Mahon et al., 2020; Sutherland et al., 2018) by improving children's self-regulation and social skills (Ezmeci & Akman, 2023; Savina, 2021; Shepley & Grisham-Brown, 2019). Self-regulation, a widely acknowledged foundational skill in early childhood, refers to the processes by which children control or direct their attention, thoughts, emotions, and actions to respond appropriately to a given situation (McClelland & Cameron Ponitz, 2012). It includes abilities such as shifting and focusing attention, ignoring irrelevant, distracting information, inhibiting impulsive behavior, or retaining instructions that help children to adjust to the demands in their daily contexts (Cameron

Ponitz et al., 2008; McClelland et al., 2007; McClelland & Cameron Ponitz, 2012). Self-regulation can thus support positive classroom climate (Khalfaoui et al., 2021), ECEC teachers' well-being, and high-quality teacher-child interactions (Ansari & Pianta, 2018; Clayback & Williford, 2022; Kwon et al., 2022). Additionally, it has been identified as a cornerstone for children's school readiness and future school success (Blair & Razza, 2007).

Despite significant influences between well-being and self-regulation (Singh & Sharma, 2018; Fomina et al., 2020), interventions and professional development opportunities that focus on ECEC teachers' well-being and children's behaviour management concertedly are still lacking (Obree et al., 2023). This goes against calls for improved support to help these professionals manage professional challenges such as chronic work overload, job strain, and under-compensation (Phillips et al., 2022). This is especially important for Portuguese ECEC teachers, since they seem to experience higher levels of job stress than their counterparts in other European countries (European Commission/EACEA/Eurydice, 2021). Indeed, in Portugal, like in other European countries, the ECEC system is facing several challenges with respect to teacher demand and supply, particularly teacher shortages and ageing (European Commission/EACEA/Eurydice, 2021). To face this, the European Commission has recently stressed the importance of making teaching careers more attractive by building on teachers' competences, offering them ongoing professional development, and by focusing on their well-being (ET2020 Working Group, 2021). In Portugal, ECEC teachers are required to hold a master's degree in early childhood and, as such, they are highly qualified compared to many other European countries (European Commission/EACEA/Eurydice, 2019). However, and despite the recent recommendations, Portuguese teachers are not regularly provided with on-site mentoring or local networking, which can affect the quality of their work (Araújo, 2018).

1.2. Integrating the PERMA Model and SWPBS Approach

Although the PERMA model and SWPBS approach are aligned on enhancing human functioning and building individual strengths in nurturing environments for the thriving of human beings (Carr & Horner, 2007; Niemiec & Ryan, 2009), interventions integrating the two into a single program to meet the needs of ECEC professionals and increase self-regulation skills of pre-school children are scarce (Piedade et al., 2024). This does not take full advantage of the potential that the combination of Positive Psychology and Positive Behaviour Support can bring towards addressing challenging behaviours and promoting positive educational experiences and personal relationships between all members of the educational community (Jenson et al., 2004).

In this paper, we address some of these shortcomings and report the results of an intervention, integrating the PERMA model and SWPBS approach, to enhance ECEC teachers' well-being levels and the self-regulation skills of children in their classrooms. The intervention included a training program that was delivered to ECEC teachers in Portugal, Greece, Cyprus, and Romania. The training program integrated elements from the PERMA model and educational strategies from the SWPBS approach into 10 sessions, which were delivered to a sample of preschool teachers during the academic years of 2021-2022 and 2022-2023.

Following previous recommendations, these sessions used a context-modeled and iterative delivery model (Walter et al., 2023) that required tailoring the structure, contents, and model of coaching to the specific needs of the Portuguese sample of ECEC teachers (see Piedade et al., 2024). The sessions were delivered on-site by two coaches who were responsible for moderating the sessions according to a practice-based coaching strategy (Sutherland et al., 2015), for monitoring the implementation of SWPBS plans, and for providing continuous support to the participating teachers (Trom & Burke, 2022). Elements of the PERMA model and strategies of the SWPBS approach were addressed in each session, with the two coaches establishing connections between the two

theoretical bases of the intervention and encouraging the participants to explore their usefulness for their daily educational practices.

In this study, we will measure the effects of the first year of the project *ProW - Promoting Teachers' Well-being through Positive Behaviour Support in Early Childhood Education* intervention on (i) Portuguese ECEC teachers' well-being levels and (ii) on the self-regulation skills of children in their classrooms. This knowledge can inform future interventions and professional development actions addressing ECEC professionals' well-being and behaviour management of preschool children.

2. Methods

2.1. Participants and procedures

To measure the longitudinal effects of the intervention, a wait-list Randomised Controlled Trial (RCT) model, with pre- and post-testing, was adopted. A total of 51 Portuguese ECEC teachers working in preschool settings located in the Municipality of Lousada (North of Portugal) participated in the intervention. All participating settings were public. Preschool in Portugal is part of the national education system for children aged between 3 and 6. Despite not being compulsory, attendance rates are relatively high at 94 %. The provision is either public or private, but all settings are under the regulation of the Ministry of Education and are required to follow the national Curriculum Guidelines for Preschool Education (European Commission/EA-CEA/Eurydice, 2019). After the Ethics Committee of the Faculty of Psychology and Educational Sciences of the University of Porto approved all ethical requirements for the study, all teachers and the legal guardians of children selected for assessment signed an informed consent for participation. The teachers were divided into eight subgroups, based on geographical location of their preschool settings. Four subgroups were randomly assigned to the experimental group (EG), while the remaining subgroups were assigned to the control group (CG).

After teachers in both groups were informed of the randomisation results, 10 children per classroom were randomly selected by the research team for teachers' behaviour assessment, and the corresponding parents or legal guardians of children in their classes were asked to consent to the indirect participation of their child by signing a consent form. Exceptions were made in the cases of teachers on official leave (Article 79), which exempted them from having a group of children under their care, and of teachers working with groups of children smaller than 10 (in these cases, teachers were asked to include all children in their groups, whenever possible). After the selection process, all participants were assigned a unique code that ensured their full anonymity in the study, and teachers in both groups were asked to respond to paper-and-pencil questionnaires at Time 1 (pre-testing, November 2021). The training sessions with teachers in the EG started shortly after. The CG followed a business-as-usual approach during the 2021–2022 academic year and received the intervention during 2022–2023 (Fig. 1).

The four subgroups in the EG included 23 teachers, with the largest subgroup including eight teachers and the smallest three teachers. These teachers attended the training program over the course of the academic year of 2021–2022. The sessions were delivered in the preschool settings

of the teachers in each subgroup. Although the sessions were scheduled considering the availability of the teachers, each group attended biweekly sessions, except at the beginning of the training program, due to restrictions imposed by the Covid-19 pandemic, and when rescheduling was specifically requested by the participants in that group. Each session had a duration of roughly an hour and a half, and integrated practical activities that were used to explore elements from the PERMA model and SWPBS approach through processes of group sharing, reflection, and feedback. For instance, in sessions 3 and 4, the importance of establishing and maintaining positive relationships for personal well-being – one of the elements of the PERMA model – was explored while also reflecting on how it can be useful in cooperating and team-working with colleagues in the same preschool setting – central to the SWPBS approach. Off-session engagement was maintained by the inclusion of activities that challenged the teachers to seek the participation of other ECEC staff, children, and their families.

Two coaches provided continuous support to the teachers in the implementation of the SWPBS plans they set during the training sessions. This support included not only sharing opportunities about off-session challenges provided at the beginning of each session, but also the availability of the coaches to schedule individual phone calls or face-to-face meetings with the participants. Further, the participants were asked to deliver records of the activities conducted (e.g., behaviour matrices developed with children). A fidelity tool was used to record data regarding attendance, engagement in the sessions, and completion of off-session activities. Further details about the contents and delivery format of the training program are described in Piedade and colleagues (2024).

After the completion of the training program, teachers in the EG and CG were asked to respond to paper-and-pencil questionnaires at Time 2 (post-testing, June 2022). For the purposes of this study, only teachers who have responded to both Time 1 and Time 2 were considered, resulting in a final sample of 30 teachers (see Table 1). The participating teachers also assessed the behaviours of a total of 397 children (see Table 2) in Time 1 and 375 children in Time 2, representing a response rate of 94.5 % from Time 1 to Time 2.

2.2. Measures

The original English versions of the two scales used to measure teachers' well-being and children's self-regulation skills were translated to standard Portuguese, using an iterative method of revising the translation of the items by each author of this paper.

2.2.1. Teachers' Well-being

Teachers' well-being was measured using the overall well-being subscale of the PERMA-Profiler (Butler & Kern, 2016) scale, a widely used tool to measure the effects of interventions on well-being (Cabrera & Donaldson, 2023; Trom & Burke, 2022). This subscale includes 16 items that assess the respondents' well-being across multiple psychosocial domains, such as positive emotions ("in general, how often do you feel joyful?"), engagement ("how often do you become absorbed in what you are doing?"), and positive relationships ("how satisfied are you with your personal relationships?"). Respondents rated each item on a Likert

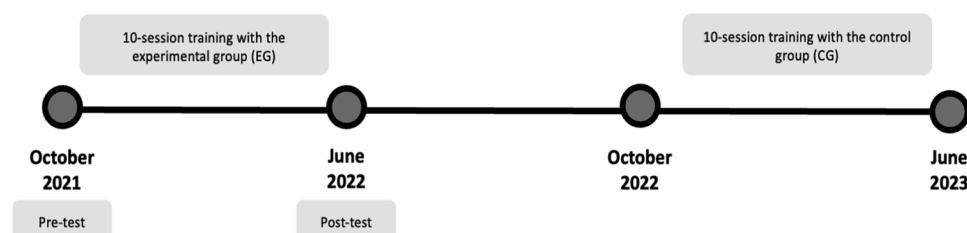


Fig. 1. Timeline of the training and data collection.

Table 1
Sociodemographic characteristics and well-being scores of teachers.

Variable	Control Group (n = 16)						Experimental Group (n = 14)					
	Time 1			Time 2			Time 1			Time 2		
	M	SD	%	M	SD	%	M	SD	%	M	SD	%
Age	51.17	5.95		52.27	6.38		53.20	6.33		54.07	6.35	
Years of teaching experience	27.86	6.63		28.86	6.63		25.29	7.12		26.29	7.12	
Years of teaching experience in the current setting	4.86	5.01		5.64	5.24		3.62	5.75		4.46	5.87	
Average number of children in class	17	4.77		18.40	4.10		19	4.05		20.21	3.40	
Highest level of education												
Bachelor			87.5			87.5			100			100
Master			12.5			12.5						
Well-being score	8.44	1.45		7.74	1.45		8.06	0.84		8.11	0.85	

Note. All the participating teachers (n = 30) were female.

Table 2
Sociodemographic characteristics and self-regulation scores of children.

Variable	Control Group (n = 217)						Experimental Group (n = 180)					
	Time 1			Time 2			Time 1			Time 2		
	M	SD	%	M	SD	%	M	SD	%	M	SD	%
Age	4.91	0.82		5.24	0.82		4.71	0.85		5.04	0.85	
Gender												
Female			50.69			50.69			46.11			46.11
Male			48.85			48.85			52.77			52.77
School Location												
Suburban			50.69			50.69			45			45
Rural			49.31			49.31			55			55
Self-regulation score	3.86	0.85		3.92	0.69		3.76	0.71		3.95	0.69	

scale, ranging from 0 (“not at all”) to 10 (“completely”). The overall well-being is given by the mean score of the 16 items, with higher scores indicating higher levels of well-being. The reliability of the overall well-being composite subscale, measured with Cronbach’s alpha (α), was high in both Time 1, $\alpha = .95$, and Time 2, $\alpha = .95$.

2.2.2. Children’s Self-regulation

Children’s self-regulation was measured using the self-regulation subscale of the Child Behaviour Rating Scale (CBRS) (Bronson et al., 1990). This measure has also been widely used to assess the effects of interventions in educational contexts (Keown et al., 2020; Lynch et al., 2004; Von Suchodoletz et al., 2013). Teachers rated specific behaviours of children in their group on a 5-point Likert scale that ranges from 1 (“never”) to 5 (“always”). We evaluated its factor structure through Confirmatory Factor Analysis (CFA) and tested for its Measurement Invariance (MI) across time and intervention conditions (i.e., experimental vs. control group). We considered an appropriate model fit when Root Mean Square Error of Approximation (RMSEA) < .06, Tucker-Lewis Index (TLI) and Comparative Fit Index (CFI) > .95, and Root Mean Residual (SRMR) < .08 (Hu & Bentler, 1999). Nested model comparison was performed through a Chi-square difference test.

Although the used version of the CBRS includes 17 items (Matthews et al., 2009), only four items were retained after conducting CFA and MI procedures. The final 4-item factor structure of the self-regulation subscale, including items 15 (“observes rules and follows directions without requiring repeated reminders”), 21 (“completes tasks successfully”), 25

(“takes time to do his/her best on a task”), and 27 (“finds and organizes materials and works in an appropriate place when activities are initiated”), showed acceptable fit to the data, $\chi^2(30) = 77.13$, $p < .001$, RMSEA = .09 (90 % CI [.06; .11]), CFI = .97, TLI = .94, SRMR = .04. Furthermore, the factor structure underlying the self-regulation scores was equivalent (scalar MI) over time and across the EG and CG (see Table 3). We further tested the reliability, through Cronbach’s alpha, of the subscale for Time 1, $\alpha = .85$, and Time 2, $\alpha = .83$.

2.3. Data analysis

To measure teachers’ well-being, we first calculated the values for the overall well-being composite subscale and then analyzed correlations to find possible associations between the teachers’ well-being scores and variables such as teachers’ age, years of experience, and number of children in their classroom. To measure the variations in the participants’ well-being between Time 1 (T1) and Time 2 (T2), we have performed analyses of variance (ANOVA) with repeated measures and main group/time interactive effects. We controlled for the potential confounding effect of teachers’ years of experience to ensure that any observed changes in well-being were not attributable to differences in teaching experience across groups. We checked ANOVA assumptions, and according to the Shapiro-Wilk test, teachers’ overall well-being scores on Time 1 ($W = 0.94$, $p = .074$) and Time 2 ($W = 0.95$, $p = .159$) did not deviate significantly from normal. We also checked for outliers, with one significant outlier being identified. The models were

Table 3
Summary of measurement invariance fit indexes: multigroup confirmatory factor analyses of the CBRS.

Models	χ^2	df	CFI	TLI	RMSEA	Compared Models	$\Delta\chi^2(df)^a$	p
Model 1: Configural	77.125	30	.964	.933	.087			
Model 2: Metric	92.100	39	.961	.944	.080	1 vs. 2	13.426(9)	.144
Model 3: Scalar	108.564	48	.957	.947	.077	2 vs. 3	15.975(9)	.067

Note. CFI = Comparative fit index; TLI = Tucker-Lewis index; RMSEA = Root-mean-square error of approximation.

^a Chi-square difference test results based on Satorra-Bentler scaling correction.

tested with and without the identified outlier, with no significant differences between the results. Therefore, we decided to keep all observations in the final model testing.

To examine self-regulation changes across the two groups, Latent Difference Score Model (LDSM) was used to estimate the latent measurement error-free change (difference scores) on self-regulation for both groups at the two time points, before and after the intervention. Specifically, we examined a multi-group LDSM using the data from the two groups, experimental and control, while controlling for child age and sex. This model assessed the intervention's effects on children's self-regulation, with its main parameters visually represented in the path diagram shown in Fig. 2.

The change in self-regulation from T1 to T2 is specified in the LDSM as a latent construct denoted as the latent difference score ($\Delta\eta_t$). This approach provides a direct estimate of average change (α_2) and its variance (ψ_2), while accounting for measurement error in the observed indicators (ϵ_i). The model's measurement component was specified using four observed self-regulation item scores (y_i), with scalar invariance constraints imposed to ensure that the measurement properties of the items (i.e., equal loadings and intercepts) remained equivalent across time points (T1 and T2) and condition (i.e., experimental and control). These constraints were applied to ensure that any observed differences in self-regulation across time or groups could be attributed to true changes in the underlying latent construct rather than changes in how the items were measured or interpreted.

The basic LDSM for two time points is given by the following general equation:

$$\eta_t = \eta_{t-1} + \Delta\eta_t$$

Where, considering the current study, η_t is the latent factor representing self-regulation at T2, η_{t-1} is the latent factor representing self-regulation at T1 and $\Delta\eta_t$ is also a latent variable representing the change between T1 and T2. The parameters of interest in the LDSM are the mean latent change (α_2) of $\Delta\eta_t$, representing the average change in self-regulation, and corresponding variance (ψ_2) of $\Delta\eta_t$, reflecting the individual variation in change around the mean. Covariates - age (mean centered) and sex (girls = 0), denoted as x_1 and x_2 in Fig. 1- were

included in the model to account for their potential influence on both the initial level of self-regulation (η_{t-1}) and the latent change score ($\Delta\eta_t$). Including these covariates ensures that the estimates of the intervention effect on self-regulation are not confounded by differences in age or sex distribution between the groups. As a result, the mean latent change (α_2) represents the expected average change in self-regulation for girls with an average age (see Newsom [2015] for further details on LDSM specification and interpretation).

For examining the differences between experimental and control groups the LDSM was estimated using a multi-group framework. This procedure provides separate estimates for the model parameters in each group, allowing for a formal test of whether the latent change ($\Delta\eta_t$) means and variances differ significantly between the experimental and control groups. The group-specific estimates were obtained by allowing the model parameters, such as latent means, variances, and structural relationships, to vary freely between groups initially. Subsequently, constraints were imposed to test for equivalence, with differences evaluated using chi-square difference testing or changes in model fit indices.

The model was tested using a hierarchically nested sample of preschoolers. Intraclass correlation coefficients (ICCs) revealed that 41 % and 53 % of the variance in child self-regulation at T1 and T2, respectively, was attributable to the classroom level. To address the classroom nesting effect, a design-based approach was employed (Muthén & Satorra, 1995). This method aggregates parameter estimates by the cluster variable (classroom) and corrects standard errors to account for non-independence of observations. To avoid excluding participants with missing data and achieve more precise model estimation, we employed full-information maximum likelihood estimation. We followed the guidelines for model fit evaluation proposed by Hu and Bentler (1999). Competing models' comparisons were performed using the likelihood-ratio test. All analyses were conducted in R (R Core Team, 2020), using the *lavaan* (Rosseel, 2012), and the *semTools* packages (Jorgensen et al., 2021).

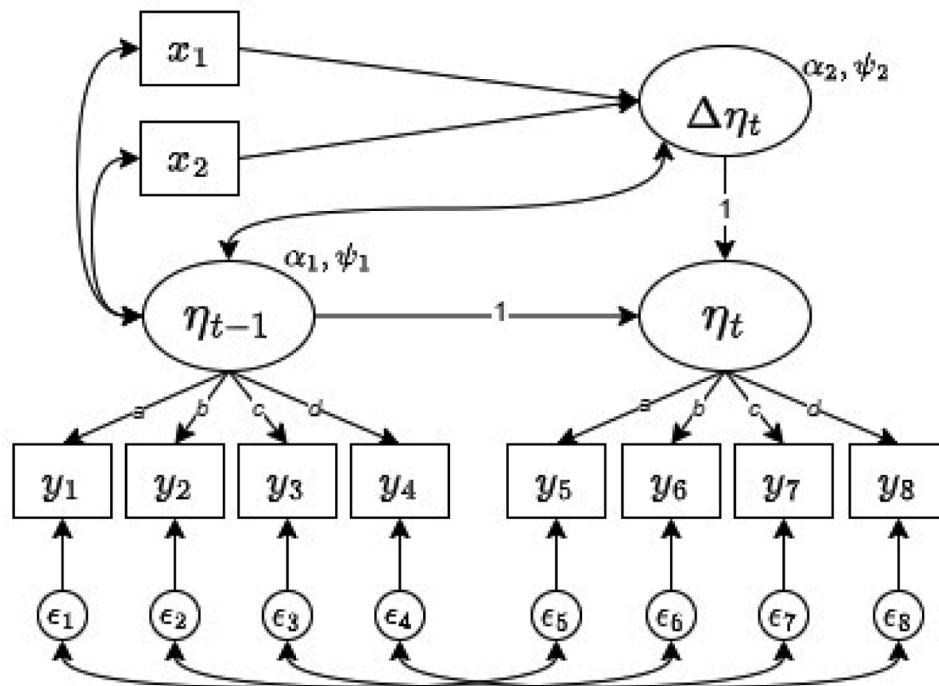


Fig. 2. Representation of the tested multigroup Latent Difference Score Model (LDSM).

Note. Observed and latent variables are depicted as rectangles and ellipses, respectively; Parameters equalized across time and dyads are denoted by letters (a – d).

3. Results

3.1. Descriptive statistics

Preliminary analysis revealed no differences between the EG and CG teachers' age ($t[25] = 1.37; p = .184$) and years of experience ($t[9] = 1.63; p = .136$). While the overall well-being scores of the EG ($n = 14$) increased from Time 1 ($M = 8.06, SD = 0.84$) to Time 2 ($M = 8.11, SD = 0.84$), the well-being scores of teachers in the CG ($n = 16$) decreased from Time 1 ($M = 8.44, SD = 1.45$) to Time 2 ($M = 7.74, SD = 1.45$). A negative correlation was found between teachers' well-being scores in Time 2 and their years of teaching experience ($r_s = -0.58, p = .04$). Furthermore, positive correlations were found between well-being scores in Time 1 and Time 2 for the EG ($r_s = 0.85, p < .001$) and the CG ($r_s = 0.74, p < .001$) (see Table 4).

The self-regulation skills of children in the EG ($n = 180$) increased from Time 1 ($M = 3.76, SD = 0.71$) to Time 2 ($M = 3.95, SD = 0.69$). A similar pattern was found for children in the CG ($n = 217$), whose self-regulation also increased from Time 1 ($M = 3.86, SD = 0.85$) to Time 2 ($M = 3.92, SD = 0.69$). Significant correlations between children's age and self-regulation skills in Time 1 ($r = 0.45, p < .001$) and Time 2 ($r = 0.38, p < .001$) were found. Self-regulation scores were also correlated between Time 1 and Time 2 for the EG ($r = 0.75, p < .001$), and for the CG ($r = 0.65, p < .001$) (see Table 5).

3.2. Intervention effects on Teachers' Well-being

A two-way repeated measures ANOVA, including the teacher's years of professional experience as a covariate, revealed no significant main effect of time on overall well-being ratings, $F(1,16) = 1.03, p = .325$, and no significant main effect of group (i.e., experimental and control, $F(1,16) = 0.09, p = .764$). Teachers' years of experience also did not significantly affect well-being ratings, $F(1,16) = 1.09, p = .311$. However, there was a significant interaction effect between group and time, $F(1,16) = 5.57, p = .031$. This indicates that changes in well-being ratings over time differed significantly between the experimental and control groups, suggesting that the intervention had a differential impact on the two groups. Bonferroni post hoc revealed that while for the EG there were no statistically significant differences between T1 and T2 ($p = .773$), for the CG, the overall well-being decreased significantly between the two times ($p = .009$). In sum, findings show a statistically significant compensatory effect of the intervention, of small effect size, $\eta^2 = 0.02$, on the well-being levels of teachers in the EG.

3.3. Intervention Effects on Children's Self-regulation

To examine the effects of the intervention on children's self-regulation, we tested a multi-group LDSM, examining for the difference between pre-testing (T1) and post-testing (T2) self-regulation scores of children in the EG and CG, while controlling for children's age and sex. As previously described, the LDSM defines a difference score factor ($\Delta\eta_t$), defined by the average difference score between self-regulation scores at T1 and T2 (denoted by α_2) and its variance (denoted by ψ_2). The mean of $\Delta\eta_t$ represents the estimated average difference

Table 4

Spearman correlations for teachers' age, years of experience, children in classrooms, and teachers' well-being ($n = 30$).

Variables	1	2	3	4	5
1. Age	-	.73	-.27	.31	-.05
2. Years of experience	.72	-	-.18	-.27	-.58*
3. Children in class	-.14	-.60	-	.19	.24
4. Perma (Time 1)	-.25	-.71	.32	-	.85**
5. Perma (Time 2)	-.01	-.31	.46	.74**	-

Note. Above Diagonal – Experimental Group; Bellow Diagonal – Control Group.

* $p < .05$ ** $p < .01$.

Table 5

Pearson correlations for children's sex, age, and self-regulation ($n = 397$).

Variables	1	2	3	4
1. Sex	-	-.12	-.12	-.17
2. Age	.09	-	.45**	.38**
3. CBRS (Time 1)	-.20*	.13*	-	.75**
4. CBRS (Time 2)	-.13	.16	.65**	-

Note. Labels for the values in the variable sex are 1 = female and 2 = male. Above Diagonal – Experimental Group; Bellow Diagonal – Control Group.

* $p < .05$ ** $p < .01$.

between the two Times - positive and negative values indicate an increase and decrease in scores from Time 1 to Time 2. The variance of $\Delta\eta_t$ can be interpreted as the random variation around the average score difference. The multigroup approach provides separate estimates for the EG and CG groups, allowing a formal test through nested model comparisons of whether the intervention significantly affected children's self-regulation.

We compared two nested models: (1) a baseline model that constrained parameters to be equal across the CG and EG, assuming no intervention effects, and (2) an alternative model that allowed for group differences by freely estimating the mean change in self-regulation (α_2) and the effects of child sex on self-regulation at T1 and the latent change score for CG and EG. The results of the model fit comparison revealed that the alternative model provided a significantly better fit than the constrained baseline model, $\Delta\chi^2(4) = 10.65, p = .031$, indicating group differences in self-regulation change. The alternative model was retained ($\chi^2[80] = 94.01, p = .136$, RMSEA = .03, 90 % CI [0.00, 0.07], CFI = .99, TLI = .99, SRMR = .08), with the estimates shown in Fig. 3.

At T1, the estimated intercept for children's self-regulation was 3.80, reflecting the average initial level of self-regulation for both the CG and the EG. Although the latent change score estimate was non-significant for the CG ($\Delta\alpha_{CG} = -0.03, p = .830$), a significant positive change in self-regulation was observed from T1 to T2 for the EG ($\Delta\alpha_{EG} = 0.20, p = .029$). Specifically, the model suggests that participation in the intervention resulted in an estimated mean of 4.0 for children's self-regulation at T2, reflecting an average increase of 0.20 from the initial level of 3.8 at T1. Although there were significant correlations between children's age, sex, and initial levels of self-regulation, none of these covariates significantly influenced the change in self-regulation between T1 and T2.

4. Discussion

In this paper, we have measured the effects of an intervention integrating the PERMA model and SWPBS approach on Portuguese ECEC teachers' well-being levels and on the self-regulation skills of children in their classrooms. Findings show that the overall teachers' well-being in the CG tended to be lower at the end of the year, whereas stable for the EG. Nevertheless, in line with previous research (Tanaka et al., 2020), our findings suggest that using the PERMA model to promote the personal and professional well-being of these professionals can have a buffering effect over the potential deterioration of their well-being levels over time. More specifically, these results may be an indication that targeting specific PERMA elements, such as Positive Emotions and Relationships, can play a protective role against the negative effects that excessive job demands and workloads (Phillips et al., 2022; Schaack & Stedron, 2020), stress (European Commission, 2021) and poor-quality relationships with other professionals and/or children in their classroom (Heilala et al., 2022) can have on the well-being of ECEC teachers over the academic year. This is also in line with research showing that positive relationships at work can act as a buffer against ECEC teachers' emotional exhaustion (Jones et al., 2020).

In fact, some of the previously mentioned factors may be at the roots of the decrease that we have found in the well-being levels of teachers,

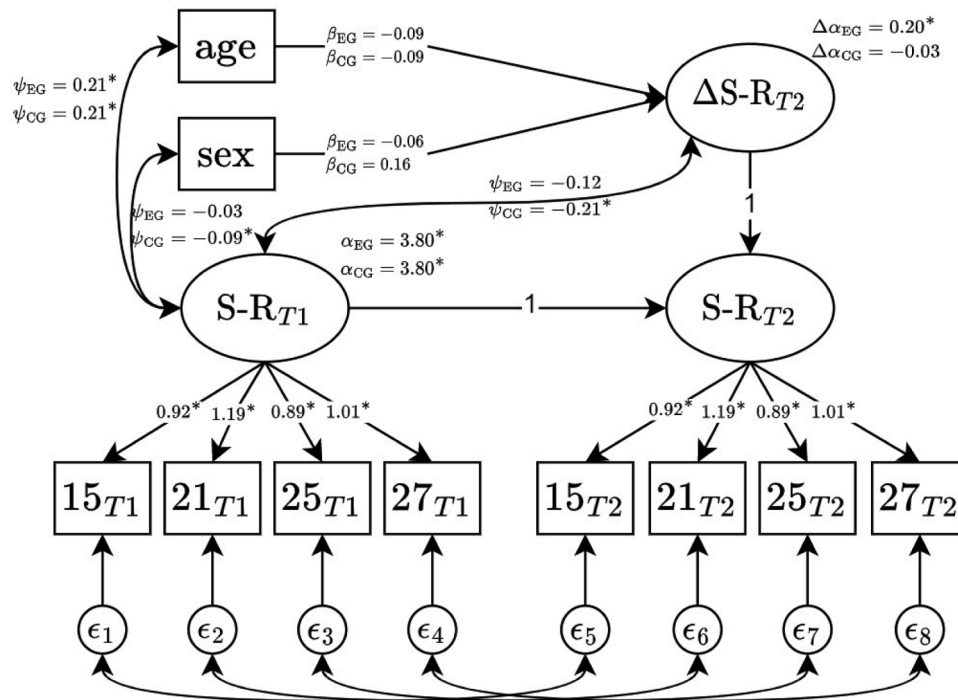


Fig. 3. Multigroup Latent Difference Score Model (LDSM) evaluating the Effects of the Intervention on Children's Self-regulation.

Note. Unstandardized coefficients for the CG and for the EG are displayed; standardized factor loadings range from .54 to .90; * $p < .05$.

both in the EG and the CG, between pre-testing (start of the academic year) and post-testing (end of the academic year). This decrease of well-being over the academic year seems to be a common trend among educators (Schaack & Stedron, 2020; von der Embse & Mankin, 2021) and is usually associated with school-specific factors, like excessive workloads and stressful work environments (Muhonen et al., 2024). The yearly accumulation of strain provoked by these factors can also be an explanation for the negative correlation that we have found between the well-being levels and years of experience of teachers in the EG. This is also in line with previous findings showing that ECEC teachers with more years of experience tend to report diminished levels of well-being (Wells, 2023).

We were also interested in assessing whether children's self-regulation skills were improved, since directly targeting the well-being of ECEC teachers can also be an important factor in creating favourable conditions for the successful implementation of SWPBS and for the consequent development of children's self-regulation. Findings show that children whose teachers participated in the program had greater improvements in their self-regulation skills, namely in observing rules, completing tasks, and organizing materials. The positive effects of the intervention on children's self-regulation skills are also in line with previous research that shows that children benefit from their teacher's participation in interventions and professional development integrating Positive Behaviour Support (Sutherland et al., 2018). These findings are important when considering the important role of self-regulation skills in early childhood on future academic achievement (Blair & Razza, 2007; Guedes & Cadima, 2022). Self-regulation can also serve as a protective factor against psychological and behavioural disorders and contribute to maintaining high well-being levels later in life (Rodríguez et al., 2022). At the social level, children's self-regulation is important in maintaining a positive classroom climate (Khalifaoui et al., 2021) and the quality of teacher-child relationships, which can ultimately support teachers' professional well-being (Ansari & Pianta, 2018).

Additionally, and despite the limitations of relying exclusively on the report of teachers to assess children's self-regulation skills, the confirmatory factor analysis, along with the measurement invariance analysis

conducted for the CBRS self-regulation subscale, resulted in a shortened version of this subscale with apparent sound psychometric properties. Our results indicate that the four items retained in the measurement invariance analysis demonstrate scalar invariance over time, making them a practical short measure of self-regulation. This tool is particularly valuable in contexts where assessing a large number of children is necessary, but resources for direct assessment or observation are limited. Besides, it offers practitioners a quick and efficient method to evaluate children's self-regulation skills and identify potential issues requiring further attention or a more thorough assessment (Fuchs & Deshler, 2007).

In combination, our results indicate that incorporating Positive Psychology and Positive Behaviour Support in interventions to promote quality ECEC can amplify the positive outcomes that these two approaches can bring, both in terms of teachers' well-being and children's self-regulation. Targeting both simultaneously can lead to a cyclical positive loop in which teachers' well-being contributes to the creation of favourable conditions for these professionals to implement quality and consistent actions to develop self-regulation skills, such as observing rules, completing tasks, and organizing materials, of children in their care. This, in turn, can help to establish quality teacher-child interactions and a positive classroom climate that may alleviate some of the negative effects that excessive workloads and stressful work environments can have on teachers' well-being. Thus, we conclude that integrating the PERMA model and SWPBS approach (Carr & Horner, 2007; Niemiec & Ryan, 2009), while adopting a context-modelled and iterative coaching model (Piedade et al., 2024; Walter et al., 2023), into future professional development actions with ECEC professionals can help to increase the quality of ECEC provision while supporting these professionals' well-being and the socioemotional development of children in their classrooms.

Nevertheless, several limitations in this study should be acknowledged. Although sampling procedures were robust, the small sample of ECEC teachers who have participated limits the generalization of these results to larger populations of ECEC professionals. Confirmatory studies, including larger samples of ECEC teachers, would be welcomed.

Another limitation concerns the use of teacher-reported data for assessing the self-regulation skills of children in their classroom. Even though the measure employed in this study is widely used and has been validated for Portuguese preschool children (e.g. Cadima et al., 2015), considering that previous studies have indicated that teachers' responses are personal, evaluative accounts of children's behaviours (Thijs & Koomen, 2009), that can be significantly influenced by contextual factors, like the nature of pre-existing teacher-child relationships (Thijs et al., 2008), our results should be viewed as representing the perspectives of the teachers, rather than the actual behaviours of children. Furthermore, despite the random selection of children in each classroom, the fact that it relied exclusively on children with consent from their parents or legal guardians may affect the representativeness of children's diverse behavioural and self-regulation profiles. Future studies measuring the effects of interventions on children's self-regulation skills should include triangulation of data collected with multiple informants, such as teachers, other ECEC staff, children's families, and external observers, using diverse data collection tools, like behaviour rating scales, but also direct observation (Hosp et al., 2003). In fact, when compared to teachers' reports or direct assessments, assessing children's self-regulation through in-context observation more effectively captures individual variations influenced by contextual features, which would enhance the robustness of the collected data (Eberhart et al., 2023). Future studies might also benefit from adopting measures that ensure representativity of diverse behavioural profiles by children in the studied sample. In addition, considering the decrease in the well-being levels of the teachers over the academic year and their negative correlation with years of experience, future research should further investigate the factors that contribute to the deterioration of these professionals' well-being over time.

5. Conclusions

Overall, this study supports the advantages of integrating the PERMA model and SWPBS approach to address ECEC teachers' well-being and preschool children's self-regulation skills simultaneously. This combination can help to amplify the positive effects of future professional development actions aiming to provide teachers with the personal and educational resources that can help them improve the quality of ECEC provided to children in their care.

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Conflict of interest statement

The authors have no conflicts of interest to declare.

CRediT authorship contribution statement

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administration, Investigation. **Joana Cadima:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Investigation, Funding acquisition, Formal analysis, Conceptualization.

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Data availability

Data will be made available on request.

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